

An Analysis of Natural language processing

Natural language processing

Natural language processing -- Part 1

1990s: Many of the notable early successes in statistical methods in NLP occurred in the field of machine translation, due especially to work at IBM Research, such as IBM alignment models. These systems were able to take advantage of existing multilingual textual corpora that had been produced by the Parliament of Canada and the European Union as a result of laws calling for the translation of all governmental proceedings into all official languages of the corresponding systems of government. However, many systems relied on corpora that were specifically developed for the tasks they were designed to perform. This reliance has been a major limitation to their broader effectiveness and continues to affect similar systems. Consequently, significant research has focused on methods for learning effectively from limited amounts of data. 2000s: With the growth of the web, increasing amounts of raw (unannotated) language data have become available since the mid-1990s. Research has thus increasingly focused on unsupervised and semi-supervised learning algorithms. Such algorithms can learn from data that has not been hand-annotated with the desired answers or using a combination of annotated and non-annotated data. Generally, this task is much more difficult than supervised learning, and typically produces less accurate results for a given amount of input data. However, large quantities of non-annotated data are available (including, among other things, the entire content of the World Wide Web), which can often make up for the worse efficiency if the algorithm used has a low enough time complexity to be practical. 2003: word n-gram model, at the time the best statistical algorithm, is outperformed by a multi-layer perceptron (with a single hidden layer and context length of several words, trained on up to 14 million words, by Bengio et al.) 2010: Tomáš Mikolov (then a PhD student at Brno University of Technology) with co-authors applied a simple recurrent neural network with a single hidden layer to language modeling, and in the following years he went on to develop Word2vec. In the 2010s, representation learning and deep neural network-style (featuring many hidden layers) machine learning methods became widespread in natural language processing. This shift gained momentum due to results showing that such techniques can achieve state-of-the-art results in many natural language tasks, e.g., in language modeling and parsing. This is increasingly important in medicine and healthcare, where NLP helps analyze notes and text in electronic health records that would otherwise be inaccessible for study when seeking to improve care or protect patient privacy.

Natural language processing -- Part 2

Where RMM is the relative measure of meaning token is any block of text, sentence, phrase or word N is the number of tokens being analyzed PMM is the probable measure of meaning based on a corpora d is the non zero location of the token along the sequence of N tokens PF is the probability function specific to a language Ties with cognitive linguistics are part of the historical heritage of NLP, but they have been less frequently addressed since the statistical turn during the 1990s. Nevertheless, approaches to develop cognitive models towards technically operationalizable frameworks have been pursued in the context of various frameworks, e.g., of cognitive grammar, functional grammar, construction grammar, computational psycholinguistics and cognitive neuroscience (e.g., ACT-R), however, with limited uptake in mainstream NLP (as measured by presence on major conferences of the ACL). More recently, ideas of cognitive NLP have been revived as an approach to achieve explainability, e.g., under the notion of "cognitive AI". Likewise, ideas of cognitive NLP are inherent to neural models multimodal NLP (although rarely made explicit) and developments in artificial intelligence, specifically tools and technologies using large language model approaches and new directions in artificial general intelligence based on the free energy principle by British neuroscientist and theoretician at University College London Karl J. Friston.

Natural language processing -- Part 3

A major drawback of statistical methods is that they require elaborate feature engineering. Since 2015, neural network-based methods have increasingly replaced traditional statistical approaches, using semantic networks and word embeddings to capture semantic properties of words. Intermediate tasks (e.g., part-of-speech tagging and dependency parsing) are not needed anymore. Neural machine translation, based on then-newly invented sequence-to-sequence transformations, made obsolete the intermediate steps, such as word alignment, previously necessary for statistical machine translation.

Natural language processing -- Part 4

Lemmatization The task of removing inflectional endings only and to return the base dictionary form of a word which is also known as a lemma. Lemmatization is another technique for reducing words to their normalized form. But in this case, the transformation actually uses a dictionary to map words to their actual form. **Morphological segmentation** Separate words into individual morphemes and identify the class of the morphemes. The difficulty of this task depends greatly on the complexity of the morphology (i.e., the structure of words) of the language being considered. English has fairly simple morphology,

An Analysis of Natural language processing

Natural language processing

especially inflectional morphology, and thus it is often possible to ignore this task entirely and simply model all possible forms of a word (e.g., "open, opens, opened, opening") as separate words. In languages such as Turkish or Meitei, a highly agglutinated Indian language, however, such an approach is not possible, as each dictionary entry has thousands of possible word forms.

Part-of-speech tagging Given a sentence, determine the part of speech (POS) for each word. Many words, especially common ones, can serve as multiple parts of speech. For example, "book" can be a noun ("the book on the table") or verb ("to book a flight"); "set" can be a noun, verb or adjective; and "out" can be any of at least five different parts of speech.

Stemming The process of reducing inflected (or sometimes derived) words to a base form (e.g., "close" will be the root for "closed", "closing", "close", "closer" etc.). Stemming yields similar results as lemmatization, but does so on grounds of rules, not a dictionary.

Natural language processing -- Part 5

$$RMM(token_N) = PMM(token_N) \times \frac{1}{2d} \left(\sum_{i=-d}^d ((PMM(token_N) \times PF(token_{N-i}, token_N, token_{N+i})) - i) \right) \quad \{\displaystyle$$
$$\{RMM(token_{\{N\}}) = \{PMM(token_{\{N\}})\} \times \frac{1}{2d} \left(\sum_{i=-d}^d ((PMM(token_{\{N\}}) \times PF(token_{\{N-i\}}, token_{\{N\}}, token_{\{N+i\}})) - i) \right) \}$$